```cpp
#include <iostream>
using namespace std;

class student {
private:   Ask ChatGPT
    int I
    string Sname;
    float Prog, Math, ImageProcessing, Total;

    float total() {
        return Prog + Math + ImageProcessing;
    }

public:
    void Takedata() {
        cout << "Enter ID: ";
        cin >> ID;
        cout << "Enter Student Name: ";
        cin >> Sname;
        cout << "Enter Programming mark: ";
        cin >> Prog;
        cout << "Enter Math mark: ";
        cin >> Math;
        cout << "Enter Image Processing mark: ";
        cin >> ImageProcessing;
        Total = total();
    }

    void Showdata() {
        cout << "\nID: " << ID
             << "\nName: " << Sname
             << "\nProgramming: " << Prog
             << "\nMath: " << Math
             << "\nImage Processing: " << ImageProcessing
             << "\nTotal: " << Total << endl;
    }
};

int main() {
    student s;
    s.Takedata();
    s.Showdata();
    return 0;
}
```

```cpp
#include <iostream>
using namespace std;

class BOOK {
private:
    int BOOK_NO;
    string BOOKTITLE;
    float PRICE;

    float TOTAL_COST(int N) {
        return PRICE * N;
    }

public:
    void INPUT() {
        cout << "Enter Book No: ";
        cin >> BOOK_NO;
        cout << "Enter Book Title: ";
        cin.ignore();
        getline(cin, BOOKTITLE);
        cout << "Enter Price per copy: ";
        cin >> PRICE;
    }

    void PURCHASE() {
        int N;
        cout << "Enter number of copies to purchase: ";
        cin >> N;
        cout << "Total cost = " << TOTAL_COST(N) << endl;
    }
};

int main() {
    BOOK b;
    b.INPUT();
    b.PURCHASE();
    return 0;
}
```

```cpp
#include <iostream>
using namespace std;

class Date {
private:
    int day, month, year;

public:
    void setDate(int d, int m, int y) {
        day = d;
        month = m;
        year = y;
    }

    void showDate() {
        cout << day << "/" << month << "/" << year << endl;
    }
};

int main() {
    Date d;
    d.setDate(7, 11, 2025);
    d.showDate();
    return 0;
}
```

```cpp
class ClassAct {
private:
    int item;
public:
    ClassAct(int x) { item = x; }
};

// إنشاء كائن ويمرر القيمة 25
ClassAct sally(25);
```

Copy code cpp

```cpp
class Circle {
private:
    double centerX, centerY, radius;
public:
    void setCenter(double, double);
    void setRadius(double);
};
```

Change (2)

❌ الأخطاء: نقص أنواع المتغيرات، وأسلوب تعريف constructor.

✅ الصحيح:

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```cpp
class Change {
private:
    int pennies, nickels, dimes, quarters;
public:
    Change() { pennies = nickels = dimes = quarters = 0; }
    Change(int p = 100, int n = 50, int d = 50, int q = 25);
};

Change::Change(int p, int n, int d, int q) {
    pennies = p;
    nickels = n;
    dimes = d;
    quarters = q;
}
```

```cpp
#include <iostream>
using namespace std;

class DumbBell {
private:
    int weight;
public:
    void setWeight(int w) { weight = w; }
    int getWeight() { return weight; }
};

int main() {
    DumbBell bar;
    bar.setWeight(200);
    cout << "The weight is " << bar.getWeight() << endl;
    return 0;
}
```

```cpp
#include <iostream>
using namespace std;

class Moon {
private:
    double earthWeight;
    double moonWeight;
public:
    Moon(double ew) {
        earthWeight = ew;
        moonWeight = earthWeight / 6;
    }
    double getMoonWeight() { return moonWeight; }
};

int main() {
    double earth;
    cout << "What is your weight? ";
    cin >> earth;
    Moon lunar(earth);
    cout << "On the moon you would weigh " << lunar.getMoonWeight() << endl;
    return 0;
}
```

```
class Date {
private:
Copy code  th, day, year;

public:
    Date(int m = 1, int d = 1, int y = 2001) {
        if (m < 1 || m > 12) m = 1;
        if (d < 1 || d > 31) d = 1;
        month = m;
        day = d;
        year = y;
    }

    void printFormat1() { cout << month << "/" << day << "/" << (year % 100) << endl; }

    void printFormat2() {
        string months[] = {"", "January", "February", "March", "April", "May", "June",
                           "July", "August", "September", "October", "November", "December"};
        cout << months[month] << " " << day << ", " << year << endl;
    }

    void printFormat3() {
        string months[] = {"", "January", "February", "March", "April", "May", "June",
                           "July", "August", "September", "October", "November", "December"};
        cout << day << " " << months[month] << " " << year << endl;
    }
};

int main() {
    Date d1(3, 15, 2013);
    d1.printFormat1();
    d1.printFormat2();
    d1.printFormat3();
    return 0;
}
```